

USB 2.0 4-Port Hub Using GL850G

1. Project Overview

This project presents the schematic and PCB design of a four-port USB 2.0 hub based on the Genesys Logic GL850G USB hub controller.

The purpose of the design is to provide multiple USB downstream ports from a single USB connection to a host computer such as a laptop or desktop PC.

The completed board contains:

- One USB upstream interface for connection to a computer
- Four USB downstream Type-A ports
- GL850G USB 2.0 hub controller
- USB power protection
- ESD protection
- Over-current protection
- External 5 V power input
- Reset circuit
- Status indication
- 12 MHz crystal oscillator
- Decoupling and filtering components
- PCB ground plane and routed USB differential signals

The GL850G supports USB 2.0 operation and can be configured for four downstream ports. Its upstream interface supports high-speed and full-speed USB operation, while its downstream interfaces support high-, full-, and low-speed USB devices.

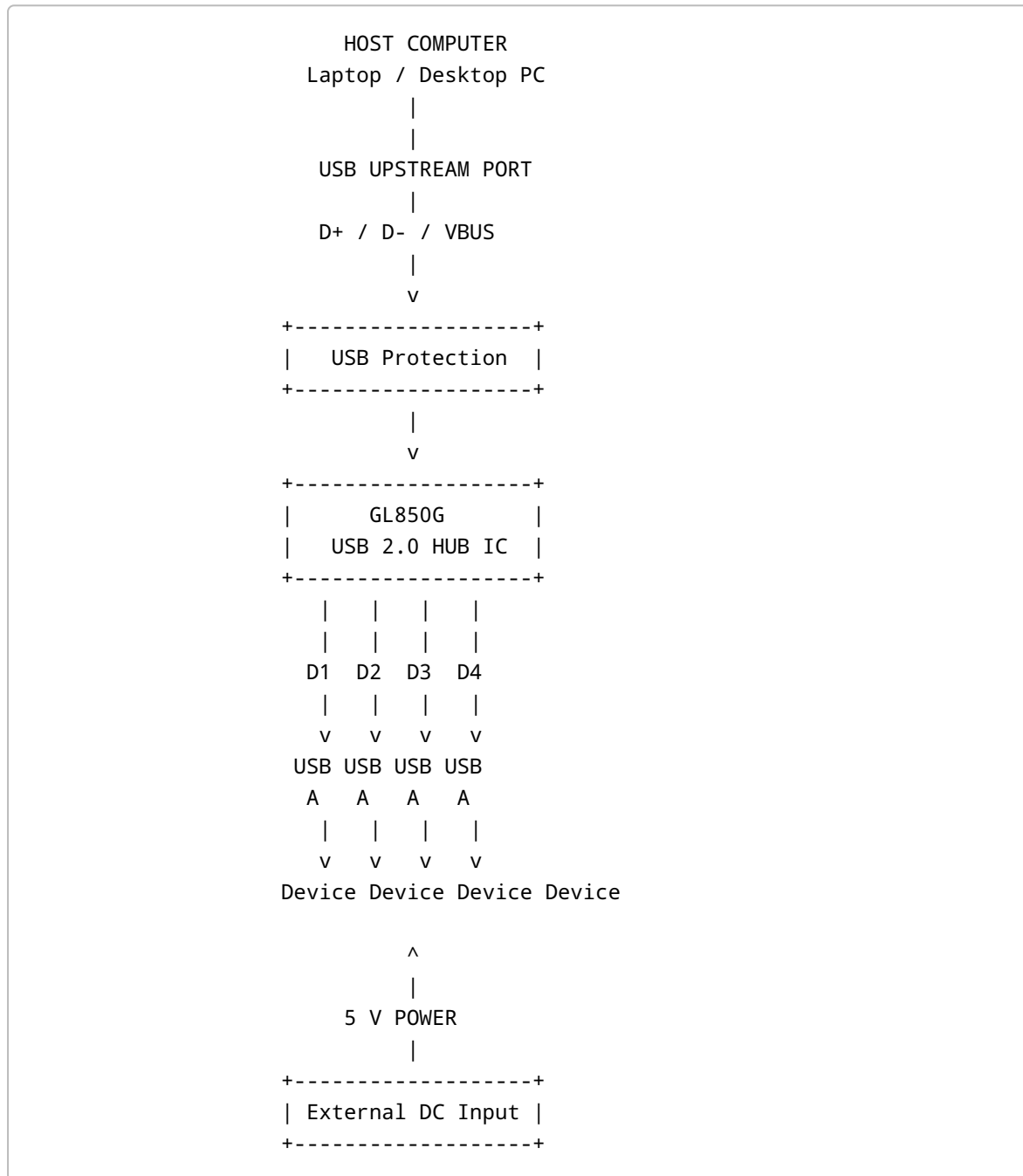
2. Project Objectives

The main objectives of this project are:

1. Design a functional four-port USB 2.0 hub.
 2. Use the GL850G as the central USB hub controller.
 3. Provide an upstream USB interface for connection to a laptop or desktop computer.
 4. Provide four downstream USB Type-A ports.
 5. Implement protection for USB power and data interfaces.
 6. Provide external power input for self-powered operation.
 7. Design the complete schematic using KiCad.
 8. Design the PCB layout using KiCad PCB Editor.
 9. Perform ERC and DRC verification.
 10. Verify the physical PCB using KiCad 3D Viewer.
 11. Generate the BOM and Gerber fabrication files.
 12. Document the complete design for future modification and fabrication.
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3. System Architecture

The overall system is organized around the GL850G USB hub controller.



The USB upstream port is the interface between the hub and the host computer. The four downstream ports are used to connect USB peripherals such as keyboards, mice, flash drives, USB adapters, and other compatible USB devices.

4. GL850G USB Hub Controller

The GL850G is the central component of the design.

It is a USB 2.0 hub controller designed for applications such as standalone USB hubs, PC motherboard hubs, notebook docking applications, and display/monitor hubs.

Important capabilities include:

- USB 2.0 compliance
- Four/three/two downstream-port configurations
- High-speed and full-speed upstream operation
- High-speed, full-speed, and low-speed downstream operation
- Integrated USB transceiver
- Single Transaction Translator (TT)
- Integrated 5 V to 3.3 V regulation
- External 12 MHz crystal/clock support
- Power management functions
- Over-current detection support
- USB signal processing and routing

The GL850G uses an internal processor and USB control logic to manage communication between the host and downstream USB devices.

USB 2.0 high-speed operation uses a nominal data rate of 480 Mb/s.

5. Upstream USB Interface

The upstream interface is the most important addition required to connect the PCB to a laptop or desktop computer.

The upstream connector should be located at the edge of the PCB so that a USB cable can be physically connected.

The basic connection is:



```

VBUS
|
|
Protection
|
+-----> Hub Power / VBUS Detection
D+
|

```

```

+-----> GL850G Upstream D+

D-
|
+-----> GL850G Upstream D-

GND
|
+-----> PCB GND

```

The upstream USB connector provides the communication path between the computer and the GL850G.

Without this connector, the four downstream USB-A connectors cannot communicate with the computer.

6. Downstream USB Ports

The design contains four USB Type-A downstream ports.

They are designated as:

```

J2 → Downstream Port 1
J3 → Downstream Port 2
J4 → Downstream Port 3
J5 → Downstream Port 4

```

Each downstream port contains:

- VBUS
- D+
- D-
- GND
- USB protection
- Power protection/current limiting where implemented

The general signal path is:

```

GL850G
|
+---- D+ / D- ---- Protection ---- USB-A
|
+---- VBUS ----- Fuse/Switch --- USB-A
|
+---- GND ----- USB-A

```

The GL850G supports four downstream ports through its port configuration.

7. USB ESD Protection

Electrostatic discharge protection is included to protect the USB interface from external electrical transients.

ESD protection should be placed close to the corresponding USB connector.

The preferred signal path is:

```
USB Connector
  |
  v
ESD Protection
  |
  v
GL850G
```

Keeping the protection device close to the connector minimizes the length of the unprotected high-speed signal path.

The same principle should be applied to the upstream USB interface and downstream USB ports.

8. Over-Current Protection

Each USB downstream port requires appropriate power protection.

The design uses protection components in the VBUS paths of the downstream ports.

Conceptually:

```
5 V Supply
  |
  v
Protection / Current Limiting
  |
  v
USB Port VBUS
```

The purpose is to reduce the possibility that a short circuit or excessive load on one USB port will damage the hub's power distribution circuitry.

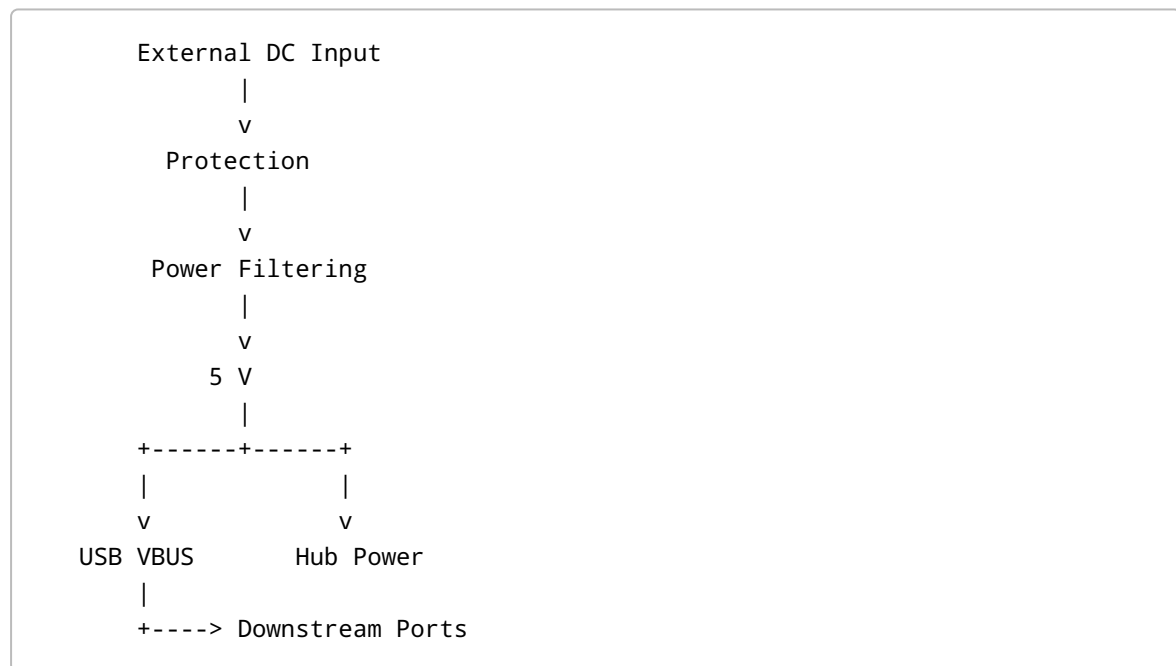
The exact current rating of the protection device should be selected according to the intended USB power architecture and the final design requirements.

9. External Power Input

The PCB includes an external DC barrel-jack power input.

This allows the hub to operate as a self-powered USB hub when external power is required.

The power architecture is approximately:



The external power input should be checked for:

- Correct input voltage
- Correct polarity
- Adequate current capacity
- Reverse-polarity protection where required
- Appropriate fuse/current limiting
- Adequate bulk capacitance

Do not connect an external supply directly to the USB host VBUS without a correctly designed power-selection/power-isolation scheme.

10. 12 MHz Crystal Oscillator

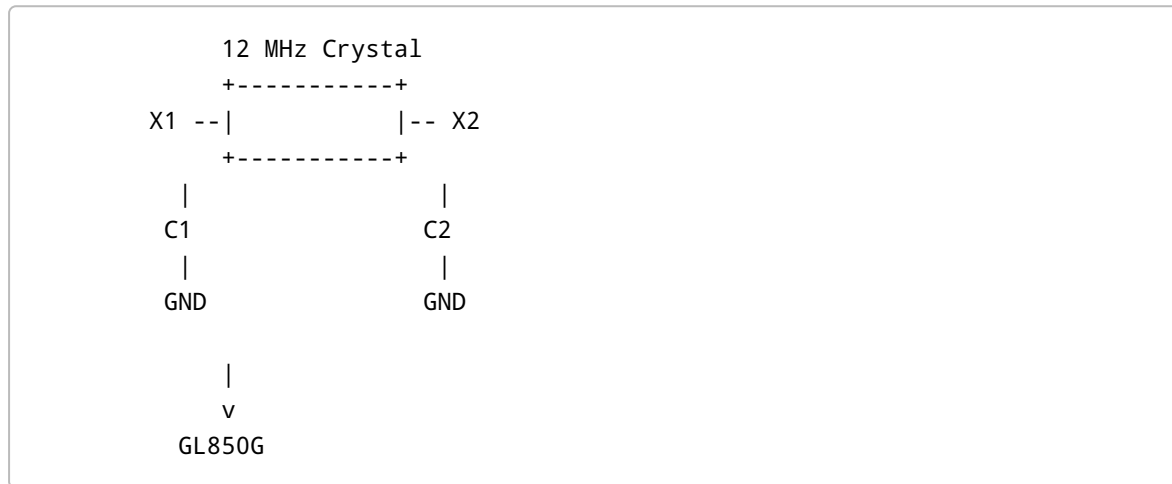
The GL850G requires an external 12 MHz clock source in the relevant configuration.

The design therefore includes:

- 12 MHz crystal
- Crystal load capacitors
- Short crystal connections

The clock section should be placed close to the GL850G.

Conceptually:



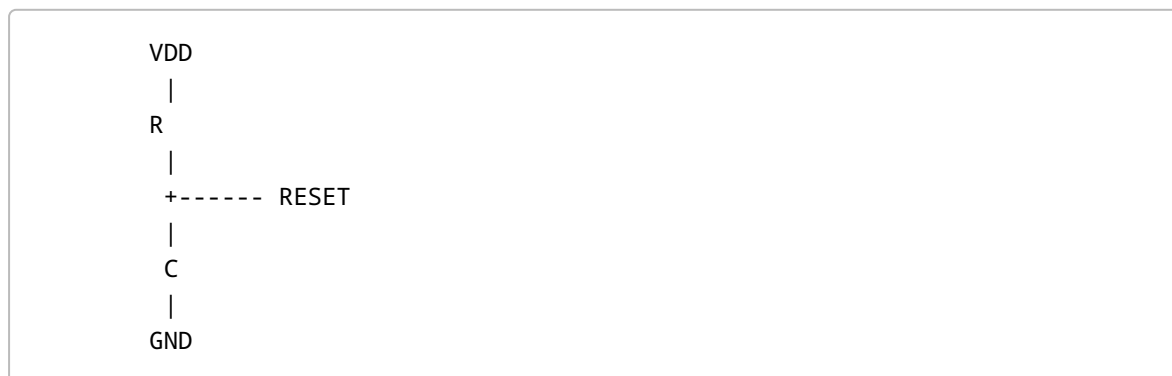
The GL850G documentation specifies support for an external 12 MHz crystal/clock source.

11. Reset Circuit

The design includes a reset circuit for the GL850G.

The reset circuit provides a defined startup condition for the controller.

A typical arrangement is:



The RC network creates a controlled reset interval during power-up.

The actual resistor and capacitor values should follow the selected GL850G package/datasheet requirements.

12. Status Indicators

LED indicators are included to provide visual information about the operating state of the board.

Possible indications include:

- Power status
- Hub activity/status
- USB port status

A typical LED circuit is:

```
VDD
|
R
|
LED
|
GND
```

The LED current-limiting resistor must be selected according to the LED characteristics and controller output capability.

13. I²C / Configuration Interface

The design includes SCL and SDA-related circuitry with pull-up resistors.

The pull-up resistors provide the required logic-high bias for the open-drain style I²C signals.

Typical arrangement:

```
VDD
|
+---- R ---- SCL
|
+---- R ---- SDA
```

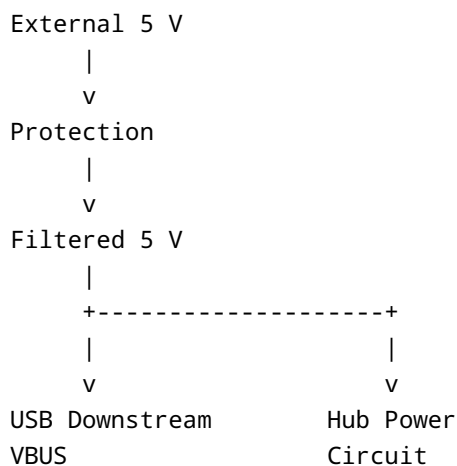
The final use of the configuration interface depends on the GL850G configuration and whether an external EEPROM or other configuration mechanism is required.

The GL850G documentation indicates that customized VID/PID and downstream-port configuration can be supported using external EEPROM configuration.

14. Power Distribution

The PCB contains a dedicated power-distribution section.

The major power paths are:



Bulk capacitors are used to reduce supply fluctuations caused by changes in load.

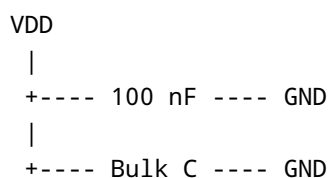
Smaller ceramic capacitors are placed close to IC power pins for high-frequency decoupling.

15. Decoupling Capacitors

Decoupling capacitors are placed near the GL850G and other active components.

Their purpose is to provide local transient current and reduce high-frequency supply noise.

Typical arrangement:



The 100 nF capacitors should be placed physically close to the relevant IC power pins.

16. PCB Layout Design

The PCB was designed using KiCad PCB Editor.

The layout includes:

- Four USB-A connectors
- Upstream USB connector
- GL850G controller
- Power input connector
- Protection components
- Crystal oscillator
- Reset circuitry
- Status LEDs
- Power filtering
- Ground plane
- USB differential-pair routing

The USB connectors are positioned around the board edges to make physical cable connection easier.

The GL850G is placed near the center of the board to reduce the distance between the controller and the USB interfaces.

17. USB Differential-Pair Routing

USB D+ and D– are high-speed differential signals.

Therefore, special attention should be given to:

- Trace length
- Differential-pair spacing
- Impedance
- Parallel routing
- Ground reference
- Avoidance of unnecessary vias
- Avoidance of sharp routing discontinuities
- Minimization of stubs

The USB 2.0 electrical compliance requirements include high-speed transmitter testing at 480 Mb/s, so the PCB layout should be treated as a high-speed design rather than ordinary low-frequency digital wiring.

The USB-IF maintains the USB 2.0 specification and electrical compliance documentation.

18. Ground Plane

A continuous ground plane should be maintained wherever practical.

The ground plane provides:

- Lower impedance return paths
- Improved signal integrity
- Reduced EMI
- Better power distribution
- Improved thermal behavior

High-speed USB signal routing should maintain a suitable continuous reference plane.

19. PCB 3D Verification

KiCad 3D Viewer was used to inspect the physical arrangement of the PCB.

The 3D inspection verifies:

- USB connector placement
- Barrel jack placement
- Component orientation
- Component clearance
- PCB dimensions
- Mechanical positioning
- Silkscreen placement

A component can appear in the 2D PCB Editor but not appear in the 3D Viewer if its footprint does not have a valid 3D model assigned.

Therefore, all important mechanical components should be checked for appropriate 3D models before fabrication.

20. Design Verification

The following verification steps should be completed before fabrication.

Schematic Verification

- Verify all component references.
- Verify component values.
- Verify power connections.
- Verify USB D+ and D- connections.
- Verify VBUS connections.
- Verify GND connections.

- Verify GL850G configuration.
- Verify crystal connections.
- Verify reset circuit.
- Verify protection circuits.

ERC

Run:

```
Schematic Editor
  ↓
Inspect
  ↓
Electrical Rules Checker
```

Resolve genuine electrical errors before proceeding.

Warnings should also be reviewed rather than blindly ignored.

21. PCB Verification

Run the KiCad Design Rules Checker.

Check:

- Unconnected nets
- Clearance violations
- Track width violations
- Via violations
- Copper-to-edge clearance
- Silkscreen-to-pad clearance
- Footprint issues
- Zone connectivity
- Differential-pair routing
- Manufacturing constraints

The final design should have no unresolved critical DRC errors.

22. Bill of Materials

After the schematic is finalized, a BOM should be generated.

Recommended BOM fields include:

Field	Description
Reference	Component reference
Quantity	Number of components
Value	Electrical/component value
Description	Component description
Footprint	PCB footprint
Manufacturer	Component manufacturer
MPN	Manufacturer part number

The BOM should be stored in:

```
BOM/
├── USB_2.0_HUB_BOM.csv
└── USB_2.0_HUB_BOM.xlsx
```

The BOM should be regenerated whenever the component list changes.

23. Gerber Generation

After the PCB has passed DRC, Gerber files should be generated for fabrication.

Typical fabrication outputs include:

- Copper layers
- Solder mask
- Silkscreen
- Board outline
- Drill files
- Other fabrication layers required by the PCB manufacturer

The generated files should be compressed into:

```
USB_2.0_HUB_Gerbers.zip
```

and stored in:

```
Gerbers/
```

24. Recommended GitHub Repository Structure

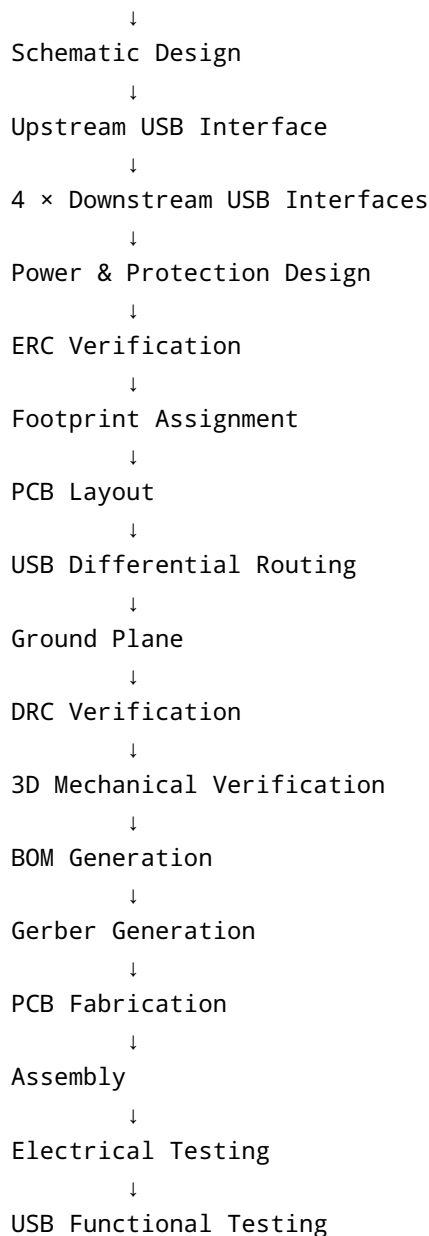
The final repository should be organized as follows:

```
USB-2.0-Hub-PCB/
|
├── README.md
├── LICENSE
├── .gitignore
|
├── KiCad/
│   ├── USB_2.0_HUB.kicad_pro
│   ├── USB_2.0_HUB.kicad_sch
│   ├── USB_2.0_HUB.kicad_pcb
│   |
│   ├── Libraries/
│   │   ├── USB_Hub.kicad_sym
│   │   └── USB_Hub.pretty/
│   |
│   └── 3D_Models/
|
├── BOM/
│   ├── USB_2.0_HUB_BOM.csv
│   └── USB_2.0_HUB_BOM.xlsx
|
├── Gerbers/
│   └── USB_2.0_HUB_Gerbers.zip
|
├── Images/
│   ├── Schematic.png
│   ├── PCB_Layout.png
│   └── PCB_3D_View.png
|
└── Documentation/
    └── USB_2.0_HUB_Design.pdf
```

25. Project Design Flow

The complete design workflow is:

```
Project Requirement
    ↓
USB Hub Architecture
    ↓
GL850G Selection
```



26. Testing Plan

After fabrication and assembly, the following tests should be performed.

Power Test

Before connecting the board to a computer:

1. Inspect the PCB for solder bridges.
2. Check resistance between power and ground.
3. Apply the external supply with current limiting.
4. Measure the 5 V rail.
5. Verify the regulated 3.3 V rail.

6. Check for excessive current consumption.

USB Host Test

Connect the upstream USB connector to a computer.

Check whether the operating system detects the USB hub.

The expected structure is:

```
Computer
|
v
USB Hub
|
+---- USB Device 1
+---- USB Device 2
+---- USB Device 3
+---- USB Device 4
```

Downstream Test

Test each port individually using known-good USB devices.

Test:

- USB flash drive
- Keyboard
- Mouse
- USB-to-serial adapter
- Other USB 2.0 compatible devices

Each downstream port should be tested separately.

27. Important Design Considerations

This project should be treated as a USB high-speed design.

Before fabrication, verify the final GL850G package, pinout, power requirements, reset configuration, crystal network, USB protection, and power architecture against the exact manufacturer documentation for the device/package being used.

This is particularly important because GL850G documentation exists for different package configurations, and some features differ between package variants.

USB-IF publishes the USB 2.0 base specification and electrical compliance documentation that should be consulted when validating a production-oriented implementation.

28. Current Project Status

Completed / Designed

- USB 2.0 hub architecture
- GL850G hub controller section
- Four downstream USB ports
- Power input section
- Protection circuitry
- Reset circuit
- Crystal oscillator section
- Status indication
- PCB layout
- PCB routing
- KiCad 3D visualization

Finalization Items

Before calling the project fabrication-ready:

- Verify the upstream USB connector and its complete protection/power connections.
- Verify the exact GL850G package and pin assignment.
- Complete ERC review.
- Complete DRC with no critical violations.
- Verify all 3D models.
- Verify USB D+/D- routing.
- Verify power-rail isolation and current protection.
- Generate the final BOM.
- Generate final Gerbers.
- Perform a final schematic-to-PCB consistency check.

29. Conclusion

The project demonstrates the complete design process of a four-port USB 2.0 hub, from schematic development through PCB layout and manufacturing preparation.

The GL850G provides the central USB hub functionality, while the PCB integrates the upstream computer interface, four downstream USB ports, power distribution, protection circuits, clock generation, reset circuitry, and status indication.

The final KiCad project files, BOM, Gerber package, PCB images, and design documentation should be maintained together in the GitHub repository so that the design can be reviewed, modified, reproduced, and fabricated in the future.

The project provides practical experience in:

- USB 2.0 hardware design

- High-speed differential routing
- Power distribution
- ESD and over-current protection
- KiCad schematic design
- PCB layout
- ERC/DRC verification
- 3D mechanical verification
- BOM preparation
- Gerber generation
- Hardware documentation
- GitHub-based hardware project management

References

1. Genesys Logic GL850G USB 2.0 Hub Controller Datasheet.
2. USB Implementers Forum – USB 2.0 Specification and Document Library.
3. USB Implementers Forum – USB 2.0 Electrical Compliance Specification.